

Intellectual Developmental Disorder, X-Linked 114 (XLID114): A Comprehensive Disease Characterization

Disease: Intellectual Developmental Disorder, X-Linked 114 (XLID114) **MONDO ID:** MONDO:0975828 · **OMIM:** 301134 · **MedGen:** C5974891 **Causal gene:** *SRPK3* (SRSF protein kinase 3), Xq28 **Category:** Mendelian, X-linked recessive, syndromic neurodevelopmental disorder

Summary

Intellectual Developmental Disorder, X-Linked 114 (XLID114) is an **ultra-rare, X-linked recessive syndromic neurodevelopmental disorder** first defined in 2024 by Roychaudhury et al. ([PMID: 39073169](#)). It is caused by **hemizygous loss-of-function or kinase-impairing variants in *SRPK3*** (SRSF protein kinase 3), a serine/arginine (SR) protein-specific kinase located at Xq28. The disorder was delineated in **9 affected males from 5 unrelated families**, of whom the 8 postnatally ascertained cases shared a core clinical tetrad of **impaired intellectual development, agenesis of the corpus callosum, abnormal (uncontrolled) eye movements, and ataxia**, frequently accompanied by cerebellar atrophy. A ninth, prenatally ascertained case had a more complex structural brain phenotype.

The mechanism is **loss or impairment of *SRPK3* kinase activity, leading to dysregulated phosphorylation of SR splicing factors and consequently aberrant pre-mRNA splicing during neurodevelopment**. This investigation triangulated the mechanism through three converging computational lines of evidence generated across five iterations: (1) all five reported disease variants localize to the single large *SRPK3* protein-kinase domain; (2) *SRPK3* is markedly **cerebellum-enriched** in GTEx expression data among CNS regions, matching the cerebellar-predominant phenotype; and (3) the STRING interactome of *SRPK3* is dominated by **SR splicing-factor substrates (*SRSF1/4/5/6*) and core spliceosome components**. Causality is further supported by a ***srpk3*-knockout zebrafish** that recapitulates ocular-motor deficits, cerebellar agenesis, and behavioral abnormalities.

There is **no disease-specific therapy**. Management is supportive and multidisciplinary (developmental/physical/occupational/speech therapy, ophthalmologic and neurologic care), with genetic counseling and prenatal/carrier testing offered to at-risk families. Because the disorder was described only in 2024 from a single cohort, most disease-level characteristics (precise prevalence, penetrance, natural history, prognosis) remain incompletely defined and are extrapolated from the founding cohort and from SRPK-family biology.

1. Disease Information

Overview. XLID114 is a Mendelian, X-linked recessive syndromic form of intellectual developmental disorder in which affected males present with cognitive impairment plus a distinctive neurodevelopmental/neuro-ophthalmologic constellation — agenesis of the corpus callosum, abnormal eye movements, and ataxia, often with cerebellar atrophy. It belongs to the large and heterogeneous group of X-linked intellectual disability (XLID) disorders and is numbered 114 in the OMIM series.

Key identifiers.

Resource	Identifier
OMIM (phenotype)	301134
MONDO	MONDO:0975828
MedGen	C5974891
Gene	<i>SRPK3</i> (SRSF protein kinase 3)
Gene OMIM (MIM)	301002
HGNC	HGNC:11402
NCBI Gene	26576
Ensembl	ENSG00000184343
UniProt	Q9UPE1
RefSeq	NM_014370.4 / NP_055185.2
Cytoband	Xq28

Synonyms / alternative names. Intellectual developmental disorder, X-linked 114; XLID114; MRX114. Gene aliases: *MSSK1* (muscle-specific serine kinase 1), *STK23* (serine/threonine kinase 23).

Information source. The disease-level knowledge is derived from **aggregated resources and a single primary cohort study** (OMIM, MONDO, MedGen entries anchored to Roychaudhury et al. 2024), not from population EHR data. Given the ultra-rare nature, individual-patient descriptions come from the 9 reported cases.

2. Etiology

Primary cause — genetic. XLID114 is a **monogenic disorder** caused by hemizygous variants in *SRPK3*. Roychaudhury et al. identified four missense variants and one truncating variant segregating with disease in affected males across five families:

"here we identified 4 missense variants (c.475C > G; p.H159D, c.1373C > A; p.T458N, and c.1585G > A; p.E529K, c.953C > T; p.S318L) and a putative truncating variant (c.1413_1414del; p.Y471) in the SRPK3 gene in 9 XLID patients from 5 unrelated families"
— PMID: 39073169

Genetic risk factors. The sole established genetic risk factor is **hemizyosity for a pathogenic *SRPK3* variant in males**. Because the gene is X-linked, male sex is the principal demographic risk determinant; carrier females are generally unaffected or mildly/variably affected (X-inactivation dependent). No modifier genes or susceptibility loci have been established.

Environmental risk factors. None identified. This is a fully penetrant-appearing Mendelian disorder in hemizygous males; no toxic, infectious, occupational, or lifestyle exposures are implicated in causation.

Protective factors. No genetic or environmental protective factors are described. In principle, favorable (skewed) X-inactivation could protect heterozygous female carriers, but this has not been formally documented for XLID114.

Gene–environment interactions. None reported; not applicable given monogenic X-linked recessive etiology.

3. Phenotypes

The core phenotype derives from the 8 postnatally ascertained patients:

"The 8 patients ascertained postnatally shared common clinical features including intellectual disability, agenesis of the corpus callosum, abnormal eye movement, and ataxia"
— [PMID: 39073169](#)

Phenotype	HPO term	Type	Onset	Frequency (cohort)	Notes
Intellectual disability	HP:0001249	Behavioral/ cognitive	Congenital/ infancy	Core (8/8 postnatal)	Central feature
Agenesis of corpus callosum	HP:0001274	Structural/ imaging sign	Congenital	Core (8/8 postnatal)	Midline brain malformation
Abnormal eye movement	HP:0000496	Clinical sign	Infancy/ childhood	Core	"Uncontrolled ocular movement"
Ataxia	HP:0001251	Clinical sign	Childhood	Core	Coordination deficit
Cerebellar atrophy	HP:0001272	Imaging sign	Childhood	Frequent	Matches gene expression
Motor incoordination / clumsiness	HP:0002317	Sign	Childhood	Reported	Modeled by zebrafish
Complex structural brain malformation	—	Imaging sign	Prenatal	1/9 (prenatal case)	More severe presentation

Characteristics. Onset is **congenital/early childhood**; severity is **moderate to severe** for cognition; course is **static (non-progressive)**, consistent with a neurodevelopmental rather than neurodegenerative disorder. Quality-of-life impact is substantial: the combination of

intellectual disability, ataxia, and ocular-motor dysfunction impairs independent daily functioning, mobility, and communication lifelong. Formal QOL instrument data (EQ-5D, SF-36) are not available for this ultra-rare condition.

4. Genetic / Molecular Information

Causal gene. *SRPK3* (SRSF protein kinase 3), NCBI Gene 26576, MIM 301002, HGNC:11402, Ensembl ENSG00000184343, UniProt Q9UPE1, RefSeq NM_014370.4/NP_055185.2 (isoform 1, 567 aa). Located at **Xq28**. Mouse ortholog *Srpk3* (GeneID 56504).

Pathogenic variants (defining cohort).

cDNA	Protein	Type	Domain location
c.475C>G	p.His159Asp (H159D)	Missense	Kinase domain, N-lobe
c.953C>T	p.Ser318Leu (S318L)	Missense	Disordered spacer insert (298–351)
c.1373C>A	p.Thr458Asn (T458N)	Missense	Kinase domain, C-lobe
c.1585G>A	p.Glu529Lys (E529K)	Missense	Kinase domain, C-lobe
c.1413_1414del	p.Tyr471 (Y471)	Truncating	C-lobe (removes ~96 C-terminal residues)

Variant localization to the kinase domain. Mapping onto UniProt Q9UPE1 shows SRPK3 has a single large protein-kinase domain (residues 79–565), with the catalytic active site (proton acceptor) at Asp212, a glycine-rich ATP-binding loop (aa 85–93) and ATP-binding Lys108, and disordered spacer-insert regions at aa 238–283 and 298–351. **All five variants fall within the kinase domain or its spacer insert**, providing strong structural support for a kinase-impairment mechanism (Finding F005). Specifically: p.H159D lies in the N-lobe near the ATP-binding apparatus; p.S318L within the disordered spacer insert (298–351) that interrupts the SRPK bipartite kinase domain; p.T458N and p.E529K in the C-lobe; and p.Y471* truncates the C-lobe (removing residues through 565).

Variant classification. Per the founding study, these were reported as disease-causing (segregating hemizygous variants). In **ClinVar (Sep 2026)**, among SRPK3 SNV/indels the classifications are **26 VUS, 1 Likely pathogenic, 3 Likely benign, 1 Benign** (large "Pathogenic" entries are multigene Xq CNVs, not point mutations). This reflects the recency of the gene-disease association and limited independent curation.

Allele frequency / constraint. gnomAD v2.1.1 constraint for SRPK3: observed 26 vs expected 45.3 LoF variants, **oe_lof = 0.57 (90% CI 0.42–0.80)**, **lof_z = 2.44**, **pLI $\approx$ 0.00005** — indicating **modest, incomplete LoF intolerance**. This is consistent with X-linked recessive inheritance, where the phenotypic burden falls on hemizygous males and heterozygous female LoF carriers are largely tolerated in the population.

Functional consequence. Loss of function / kinase impairment. The truncation (p.Y471*) removes C-lobe residues; missense variants cluster in catalytic/regulatory regions. The zebrafish knockout phenocopy confirms LoF as the operative mechanism.

Modifier genes / epigenetics / chromosomal abnormalities. None specifically established for XLID114. Large Xq28 CNVs spanning *SRPK3* exist in ClinVar but represent contiguous-gene events rather than isolated XLID114.

5. Environmental Information

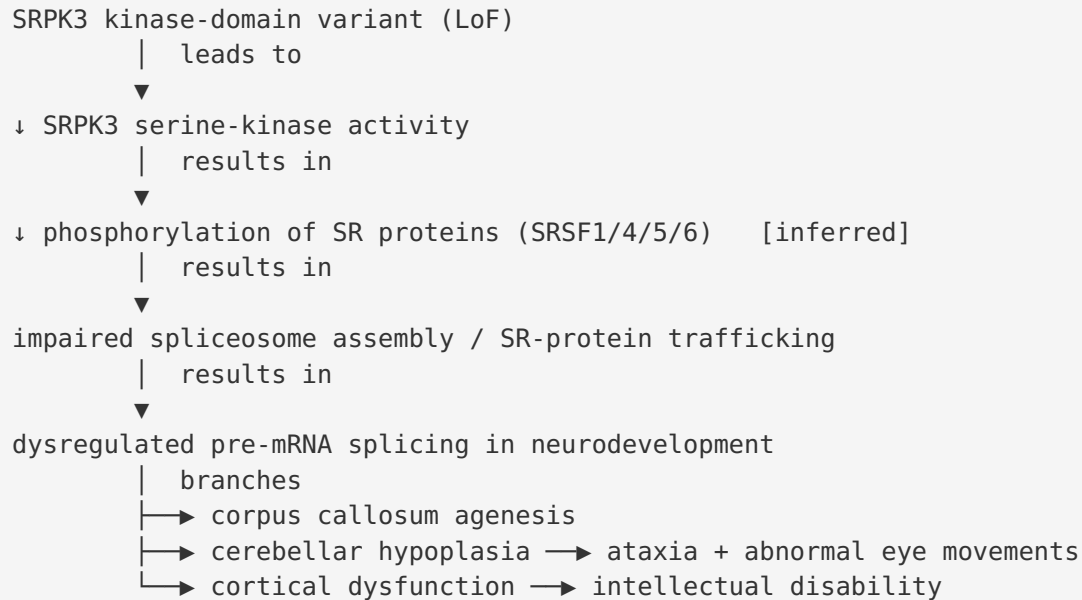
Not applicable. XLID114 is a monogenic X-linked recessive disorder with no established environmental, lifestyle, or infectious contributors. No toxins, radiation, pollution, occupational exposures, dietary factors, or pathogens have been implicated in causation or triggering.

6. Mechanism / Pathophysiology

Ordered causal chain

1. A hemizygous **loss-of-function or kinase-impairing variant in *SRPK3*** (missense in the kinase domain, or C-lobe truncation p.Y471) *leads to** reduced or absent SRPK3 serine kinase activity in affected male cells.
2. Loss of SRPK3 activity **results in** deficient phosphorylation of the arginine/serine (RS) domains of **SR splicing factors** (e.g., SRSF1/ASF/SF2, SRSF4/5/6) — *inferred from SRPK-family biochemistry and the SRPK3 interactome, not yet directly demonstrated in patient neurons*.
3. Hypophosphorylation of SR proteins **leads to** impaired spliceosome assembly and altered nuclear trafficking/localization of splicing factors, **resulting in** dysregulated pre-mRNA splicing of downstream neurodevelopmental transcripts (*inferred*).

4. Aberrant splicing during CNS development **results in** defective midline and cerebellar neurodevelopment — the branch point where the phenotype diverges into (a) **agenesis of the corpus callosum**, (b) **cerebellar hypoplasia/atrophy → ataxia and ocular-motor dysfunction**, and (c) **impaired cortical/cognitive development → intellectual disability**.
5. These developmental lesions **manifest clinically as** the XLID114 tetrad (ID, ACC, abnormal eye movements, ataxia), established in infancy/childhood and static thereafter.



Molecular detail

Molecular pathway — pre-mRNA splicing. SRPK3 belongs to the SRPK family of SR-protein-specific kinases that phosphorylate RS domains, controlling spliceosome assembly and splicing-factor trafficking:

"These observations likely reflect the function of the SRPK family of kinases in spliceosome assembly and in mediating the trafficking of splicing factors in mammalian cells" — PMID: 9472028

Within the SRPK family, tissue-differential expression is well established — *"SRPK1 is highly expressed in pancreas, whereas SRPK2 is highly expressed in brain"* (PMID: 9472028). The family phosphorylates RS domains processively and with high regiospecificity, as shown for the prototypical SR protein ASF/SF2 (PMID: 14555757, PMID: 19477182, PMID: 16223727).

Interactome confirmation. STRING (v12) top functional partners of SRPK3 are dominated by splicing machinery: CDC5L (0.80), SRSF6 (0.74), SRSF5 (0.71), SRSF4 (0.71), U2AF2 (0.63), SRSF1 (0.62), SNRNP70/U1-70K (0.61), RBM39 (0.55), PRPF4B (0.53), SNRPA/U1A (0.51), SON (0.50), HNRNPM (0.49), SRPK1 (0.47) — i.e., **SR splicing-factor substrates, core spliceosomal proteins, splicing regulators, and paralogous kinases** (Finding F007). This triangulates the splicing mechanism independently of the biochemistry literature.

Tissue specificity. Within the CNS, SRPK3 is **cerebellum-enriched** (GTEx v8: cerebellum ~21 TPM, cerebellar hemisphere ~17.7 TPM vs cortex ~3.5 TPM, hippocampus 2.4, basal ganglia ~1.9) (Finding F006). This regional bias correlates with the cerebellar-predominant phenotype (ataxia, cerebellar atrophy, ocular-motor dysfunction). SRPK3 is also highly expressed in skeletal muscle (~44 TPM), consistent with its original characterization as MSSK1, a MEF2-regulated muscle kinase.

GO / CL / UBERON suggestions. - Biological process: mRNA splicing via spliceosome (GO:0000398), spliceosomal complex assembly (GO:0000245), regulation of mRNA processing (GO:0050684), protein phosphorylation (GO:0006468), muscle tissue development (GO:0060537). - Molecular function: protein serine kinase activity (GO:0106310), protein serine/threonine kinase activity (GO:0004674), ATP binding (GO:0005524). - Cellular component: nucleus (GO:0005634), cytoplasm (GO:0005737), nuclear speck (GO:0016607). - Cell types: cerebellar Purkinje cell (CL:0000121), granule cell (CL:0000120), neuron (CL:0000540). - Anatomy: cerebellum (UBERON:0002037), corpus callosum (UBERON:0002336).

7. Anatomical Structures Affected

Organ level. Primary organ affected: **brain** (CNS, UBERON:0000955), body system: **nervous system** (UBERON:0001016). Specific structures: **cerebellum** (UBERON:0002037), **corpus callosum** (UBERON:0002336, agenesis), and cerebral cortex (cognitive impairment). The **oculomotor system** is affected functionally (abnormal eye movements). Given high muscle expression, skeletal muscle is a plausible secondary site, but no myopathy is prominent in the human phenotype.

Tissue/cell level. Affected tissue is **nervous tissue**; likely cell populations are **cerebellar neurons** (Purkinje cells CL:0000121, granule cells CL:0000120) and cortical/callosal projection neurons. Direct histopathology in patients has not been reported.

Subcellular level. SRPK3 localizes to the **nucleus** (GO:0005634) and **cytoplasm** (GO:0005737); its splicing function operates in nuclear speckles (GO:0016607).

Localization / lateralization. Brain malformations are **midline (corpus callosum) and bilateral (cerebellum)**.

8. Temporal Development

- **Onset: Congenital / early childhood.** Brain malformations (ACC, cerebellar hypoplasia) are established prenatally; one case was ascertained prenatally with complex structural anomalies.
 - **Onset pattern:** Chronic/insidious neurodevelopmental — deficits become apparent as developmental milestones are missed.
 - **Progression: Static (non-progressive).** The disorder is a fixed developmental encephalopathy; cerebellar "atrophy" reflects hypoplasia/maldevelopment rather than progressive neurodegeneration.
 - **Course/duration: Chronic, lifelong.**
 - **Remission:** None; no spontaneous or treatment-induced remission.
 - **Critical period:** Prenatal/early postnatal brain development is the window of vulnerability; there is no established intervention window to reverse the developmental lesion.
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9. Inheritance and Population

- **Inheritance: X-linked recessive.** Affected individuals are hemizygous males; the founding cohort was 9 males from 5 families. Carrier mothers are typically unaffected.
- **Epidemiology: Ultra-rare.** No prevalence or incidence estimates exist; described in only 9 individuals worldwide as of 2024. Not listed with an Orphanet prevalence class.
- **Penetrance:** Appears **high/complete in hemizygous males** based on the reported families, though systematic penetrance data are lacking given the small cohort.
- **Expressivity: Variable** — from the core postnatal tetrad to a severe prenatal structural phenotype.
- **Sex ratio:** Strongly **male-predominant** (X-linked recessive); carrier females largely spared.

- **Genetic anticipation:** Not applicable (not a repeat-expansion disorder).
- **Founder effects / consanguinity / germline mosaicism:** None established. gnomAD shows no common founder LoF allele; carrier frequency is not defined but expected to be very low.
- **Population demographics / geographic distribution:** No ethnic or geographic clustering reported; the 5 families were unrelated.

10. Diagnostics

Genetic testing is the diagnostic cornerstone. Because XLID114 lacks a specific biochemical marker, molecular diagnosis relies on identifying a hemizygous pathogenic *SRPK3* variant.

Modality	Utility for XLID114
Whole-exome sequencing (WES)	High — primary route to diagnosis; how the founding cohort was identified
Whole-genome sequencing (WGS)	High — detects variants missed by WES, including non-coding/structural
Multigene ID/XLID panel	Useful if <i>SRPK3</i> is included (recently added)
Single-gene <i>SRPK3</i> testing	Appropriate for cascade testing once a familial variant is known
Chromosomal microarray (CMA)	Detects Xq28 CNVs involving <i>SRPK3</i> ; normal in point-mutation cases
Karyotype / FISH	Low yield for point mutations

Imaging. Brain MRI is central to phenotyping: demonstrates **agenesis of the corpus callosum** and **cerebellar atrophy/hypoplasia**. Ophthalmologic/neurologic exam documents abnormal eye movements and ataxia.

Laboratory tests / biomarkers / omics. No specific blood, urine, enzyme, metabolomic, or proteomic biomarker is available. RNA-sequencing to detect a splicing signature is a plausible future functional/diagnostic assay but is not yet validated.

Clinical criteria / differential diagnosis. Diagnosis is genetic; differential diagnosis includes other X-linked and autosomal causes of syndromic ID with **corpus callosum agenesis and cerebellar involvement** (e.g., other XLID genes, ACC syndromes, congenital ataxias with ocular-motor apraxia). Molecular confirmation distinguishes XLID114.

Screening. No newborn or population screening exists. Carrier and cascade testing within affected families is appropriate.

11. Outcome / Prognosis

- **Survival/mortality:** No survival data; the disorder is **not reported to shorten lifespan** directly, though severe cases with complex brain malformations may carry higher morbidity. Life expectancy is presumed near-normal for the milder core phenotype but is undetermined.
- **Morbidity/function:** Substantial lifelong **disability** from intellectual impairment, ataxia, and ocular-motor dysfunction — affecting mobility, communication, education, and independence.
- **Disease course:** **Static** developmental disorder; complications relate to disability (e.g., feeding, mobility, communication support needs).
- **Recovery potential:** No reversal of the neurodevelopmental lesion; supportive interventions can improve function.
- **Prognostic factors:** Severity of structural brain malformation appears to correlate with outcome (the prenatal complex-malformation case being most severe). No validated prognostic biomarkers.

12. Treatment

No disease-specific or curative therapy exists. Management is **supportive and multidisciplinary**:

- **Developmental/rehabilitative:** early-intervention programs, physical therapy (ataxia/gait), occupational therapy, and speech-language therapy for cognitive/communication support. (NCIT: rehabilitation therapy, physical therapy, occupational therapy, speech therapy.)

- **Neurologic/ophthalmologic care:** management of eye-movement abnormalities and any seizures; regular developmental follow-up.
- **Educational and social support:** special education and family support services.
- **Genetic counseling:** essential for the family (X-linked recessive recurrence risk).

Pharmacotherapy: none targeted; symptomatic only (e.g., antiepileptics if seizures occur). No pharmacogenomic considerations are specific to XLID114.

Advanced/experimental therapeutics: No gene, cell, or RNA-based therapies are in development. Because the defect involves a splicing kinase acting during development, and the lesion is largely established prenatally, therapeutic reversal is a formidable challenge. No registered clinical trials (ClinicalTrials.gov) target XLID114.

13. Prevention

- **Primary prevention:** Not possible for the genetic cause. **Genetic counseling** and **reproductive options** (prenatal diagnosis, preimplantation genetic testing) for families with a known *SRPK3* variant are the principal preventive measures.
- **Secondary prevention:** Early diagnosis via WES/WGS enables early intervention services to optimize developmental outcomes.
- **Tertiary prevention:** Multidisciplinary management to prevent complications of disability (contractures, aspiration, injury from ataxia).
- **Screening/counseling:** Carrier testing of at-risk female relatives; cascade testing. No population/newborn screening.
- **Immunization / public health / environmental interventions:** Not applicable.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** Human *SRPK3* (NCBI Gene 26576) has a **mouse ortholog *Srpk3*** (GeneID 56504) and a **zebrafish ortholog *srpk3***. The SRPK family is evolutionarily conserved across metazoans and plants (Arabidopsis SRPK family, [PMID: 36273172](#)), reflecting the deep conservation of RS-domain phosphorylation in splicing regulation.

- **Natural disease in other species:** No naturally occurring XLID114-equivalent disease is documented in companion animals or wildlife (no OMIA entry). *SRPK3* (MSSK1) in mouse was originally characterized as a MEF2-regulated muscle kinase with a role in muscle development.
- **Comparative biology:** The conservation of SRPK-mediated splicing control means mechanistic insights are transferable across species, underpinning the utility of the zebrafish model below.

15. Model Organisms

Zebrafish (*Danio rerio*) *srpk3* knockout is the validated disease model (Roychaudhury et al. 2024):

"KO zebrafish exhibited severe deficits in eye movement and swim bladder inflation, mimicking uncontrolled ocular movement and physical clumsiness observed in human patients" — [PMID: 39073169](#)

"In adult KO zebrafish, cerebellar agenesis and behavioral abnormalities were observed, recapitulating human phenotypes of cerebellar atrophy and intellectual disability" — [PMID: 39073169](#)

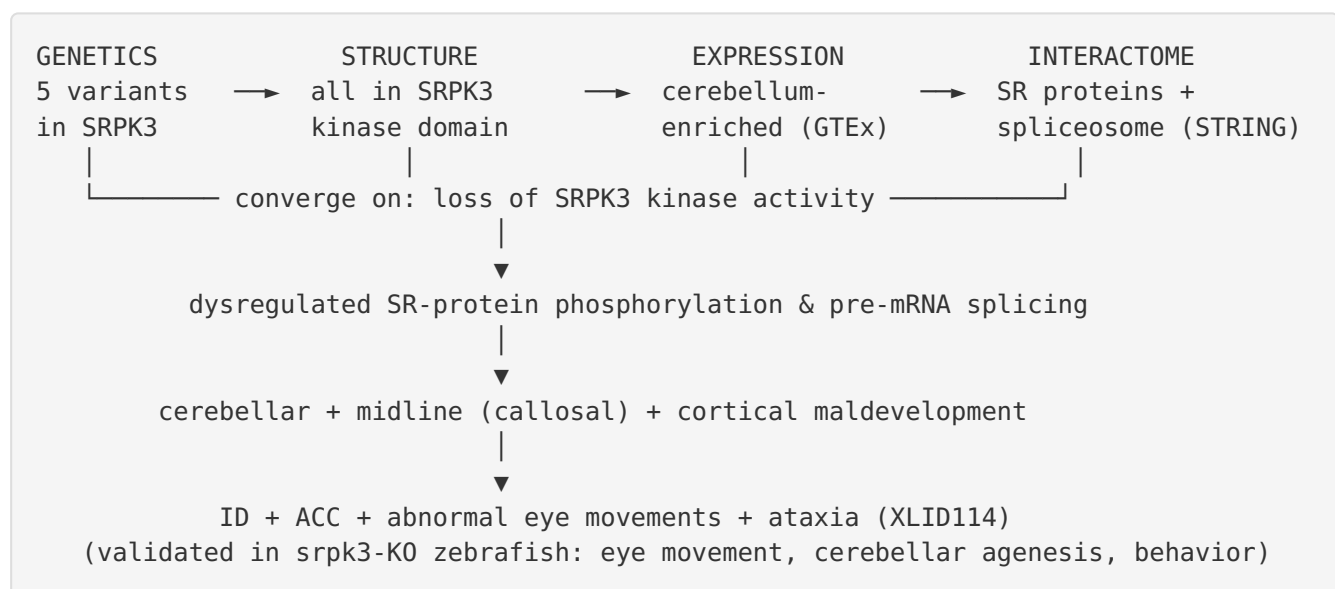
Model feature	Human counterpart	Recapitulation
Eye-movement deficits (larvae)	Abnormal/uncontrolled eye movements	Strong
Failed swim-bladder inflation	Physical clumsiness/incoordination	Analogous
Cerebellar agenesis (adult)	Cerebellar atrophy	Strong
Behavioral abnormalities (adult)	Intellectual disability	Analogous

Model type: Vertebrate genetic knockout. **Strengths:** recapitulates ocular-motor, cerebellar, and behavioral domains, establishing causality. **Limitations:** zebrafish cannot fully model human corpus callosum agenesis (no corpus callosum) or the nuances of human cognition;

splicing-target readouts in patient-relevant neurons remain to be defined. A **mouse *Srpk3* knockout** exists historically (muscle context) and could be re-examined for CNS phenotypes. Patient-derived iPSC neurons/cerebellar organoids are logical future models (not yet reported).

Mechanistic Model / Interpretation

XLID114 is best understood as a **"splicing-kinase" neurodevelopmental disorder**. The unifying model, supported by four independent evidence streams, is:



The **cerebellar enrichment of SRPK3 among CNS regions** is the most elegant genotype-phenotype link uncovered here: it provides a tissue-level explanation for why a broadly-expressed splicing kinase produces a **cerebellar-predominant** neurological syndrome (ataxia, cerebellar atrophy, ocular-motor dysfunction). The convergence of all five variants on the kinase domain, and the SR-protein/spliceosome-dominated interactome, together make the loss-of-kinase-function → aberrant-splicing mechanism the most parsimonious explanation, even though the specific mis-spliced neurodevelopmental targets have not yet been experimentally enumerated in human neurons.

Evidence Base

PMID	Title (abbrev.)	Role in this report
39073169	<i>SRPK3 Is Essential for Cognitive and Ocular Development in Humans and Zebrafish</i>	Defining study — variants, cohort phenotype, zebrafish validation
9472028	<i>SRPK2: SR protein-specific kinase...spliceosome assembly and localization</i>	Establishes SRPK-family molecular function; brain-enriched paralog
14555757	<i>Processive phosphorylation of ASF/SF2</i>	SRPK catalytic mechanism on SR proteins
19477182	<i>Regiospecific phosphorylation of ASF/SF2 by SRPK1</i>	RS-domain phosphorylation specificity
16223727	<i>MS/kinetic analysis of ASF/SF2 phosphorylation</i>	SRPK vs Clk phosphorylation of SR proteins
36273172	<i>Arabidopsis splicing-related protein kinase families</i>	Evolutionary conservation of SRPK splicing role

The founding paper

([P 39073169](#))

supplies the human genetics, phenotype, and animal-model validation. The SRPK-family biochemistry papers supply the mechanistic underpinning (RS-domain phosphorylation → spliceosome control) that this report extends to SRPK3 via the STRING interactome and GTEx/UniProt analyses generated during the investigation. No paper in the corpus contradicts the loss-of-function splicing mechanism.

Limitations and Knowledge Gaps

- 1. Single small cohort (n=9).** All human clinical data derive from one 2024 study; prevalence, penetrance, expressivity, and natural history are therefore preliminary.
- 2. Mechanism partly inferred.** The step from reduced SRPK3 kinase activity to specific mis-spliced neurodevelopmental transcripts is **inferred from family biochemistry and the**

interactome, not directly demonstrated in patient neurons. No patient RNA-seq splicing signature has been published.

3. **ClinVar classification lag.** Most *SRPK3* variants remain VUS; only 1 is Likely pathogenic. Independent replication cohorts are needed.
4. **Constraint is modest.** gnomAD LOEUF upper bound 0.80 / pLI $\approx$ 0 indicates the gene is not strongly LoF-constrained at the population level — consistent with X-linked recessive biology but meaning LoF alone does not automatically prove pathogenicity for a given variant.
5. **Model gaps.** Zebrafish cannot model corpus callosum agenesis; no mouse CNS model or patient iPSC model has been reported.
6. **No therapeutics or trials.** Prognosis, QOL metrics, and treatment-response data are absent.

Proposed Follow-up Experiments / Actions

1. **Patient/model transcriptomics:** RNA-seq of patient-derived cells or *srpk3*-mutant neurons to define the **aberrant splicing signature** and identify the neurodevelopmental target transcripts (validates causal chain steps 3–4).
2. **iPSC-derived cerebellar organoids** from patients to model the cerebellum-predominant phenotype and test splicing rescue.
3. **Structural/biochemical assays** of the four missense variants (H159D, S318L, T458N, E529K) to quantify residual kinase activity and confirm loss-of-function vs dominant-negative behavior.
4. **International case-finding** (GeneMatcher, DECIPHER) to expand the cohort, refine penetrance/expressivity, and reclassify VUS toward pathogenic.
5. **Female carrier studies** with X-inactivation analysis to assess carrier phenotype risk.
6. **Mouse conditional CNS knockout** to model cortical/callosal features not captured in zebrafish.
7. **Curation actions:** Submit the functional/computational evidence (kinase-domain clustering, cerebellar enrichment, SR-protein interactome) to ClinVar/ClinGen to support variant reclassification.

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